Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Arctic – Monitoring Forecast Centre

ARCTIC_ANALYSISFORECAST_PHY_ICE_002_011

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CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The first column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	June 2020	All	Creation of the document	A. Melsom	L. Bertino, ARC MFC leader
1.1	September 2020	I, IV, V	Updated model code, new analysis period	A. Melsom, T. Williams	L. Bertino, ARC MFC leader
1.2	September 2021	I, IV, V	Updated model code, new analysis period	A. Melsom, T. Williams	L. Bertino, ARC MFC leader
1.3	September 2022	All	Added validation of results for sea ice type, surface albedo. Reprocessed validation with results from model version upgrade.	A. Melsom, T. Williams	L. Bertino, ARC MFC leader
1.4	August 2023	All	Updated model code, added validation of results for concentration of young ice and volume fraction of young ice; expanded sections with results for sea ice thickness and sea ice type/age.	A. Melsom, T. Williams	L. Bertino, ARC MFC leader

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

This document covers the following product:

- ARCTIC_ANALYSISFORECAST_PHY_ICE_002_011, which is one of the two real-time regional forecasts of physical sea ice parameters in the Arctic.

The product includes one 2D hourly averaged dataset called cmems_mod_arc_phy_anfc_nextsim_hm interpolated onto a 3 km polar stereographic output grid. The sea ice concentrations (total, multi-year and young), thickness, drift velocity, ice type, albedo and snow depths are provided. The model domain covers the central Arctic region (see Section III), excluding the Canadian Archipelago and the Pacific side of Bering Strait, which is smaller than the area of responsibility of the ARC MFC, but is planned to be completed in a further upgrade of the product.

This product serves as an advanced supplement to the coupled ocean-sea ice forecasts from ARC MFC, which presently is the ARCTIC_ANALYSISFORECAST_PHY_002_001.

For the present qualification, all metrics (estimated accuracy numbers - EANs) were computed on the output grid, as available to Copernicus Marine Service (CMEMS) catalogue users.

More technical details on the products mentioned here are available in the Product User Manual (PUM).

I.2 Summary of the results

Validation of products from the operational forecasts has been performed using remote sensing data sets for sea ice concentration (SIC), sea ice drift and thickness of thin sea ice.

Based on the results provided in Section IV, we deem the results from this Arctic MFC product to be of satisfactory quality. However, there are some obvious areas in which improvements in the quality of products should be sought:

- Except around the time of the sea ice minimum, the marginal ice zone in the model results is too small.
- The IIEE (integrated ice edge extent) area has a negative bias during periods where the ice edge length is overestimated, so lower sea ice extent in the model results is associated with an excessively undulating character of the sea ice edge in the model results.
- The sea ice thickness (SIT) in regions with thin sea ice is too large, particularly around the time of the winter ice maximum.
- Some arching forms between North-east Greenland and Svalbard, resulting in sea ice export through the Fram Strait being too low in summer. The summer sea ice extent in the Greenland Sea is accordingly too low as well.
- Early in the year, there is a positive bias in the first year ice area, and a corresponding negative bias for the multi-year ice. Approaching summer, this situation reverses.
- The sea ice age has a root mean square difference (RMSD) of about 110 days, with a variable bias that is zero on average.
- The model is consistently overestimating surface albedo, with a bias ranging from 0 to 0.1 and RMSD ranging from 0.1 to 0.175.

We have adopted the definition of *sea ice area* that is commonly used, which is the areal integral of the sea ice concentration. In contrast, the *sea ice extent* is the area bounded by one sea ice concentration contour (the 0.15 concentration contour is frequently used), or between a pair of contours for different sea ice concentration values.

I.3 Estimated Accuracy Numbers

Results in Table 1 are based on analyses of model results from weekly forecast bulletins. This is a summary of results from Tables 2-11 in Section IV.1-7. With the exceptions of sea ice age and surface albedo (see footnotes for details), model results were compared to observations from the period 2019-10-16 – 2020-10-01. Bias (mean difference) is calculated by subtracting observations from model results (i.e., model value – observed value).

Table 1: *Estimated accuracy numbers. Root-mean-square difference (RMSD) and mean absolute difference (MAD) for various parameters. Generally, observations from 2019-10-16 – 2020-10-01 were used. However, observations of sea ice thickness and ice type (first year ice, multiyear ice) are not available during summer. The periods with data for sea ice age and surface albedo are also exceptions, see the relevant footmarks. Details about the supporting observations are given in Section III.2. Columns "FC day 1" and "FC day 6" correspond to results for the 1st and 6th day relative to the forecasts' bulletin dates.*

Variable	Region	Supporting observations	FC day 1	FC day 6	FC day 1	FC day 6
			MAD		Bias (model-observation)	
Sea ice concentration	Full domain	SSMIS	0.11	0.12	0.07	0.05
MIZ[1] ice area			0.69 Mkm^2	0.79 Mkm^2	-0.62 Mkm^2	-0.74 Mkm^2
MIZ extent			1.02 Mkm^2	1.23 Mkm^2	-0.95 Mkm^2	-1.20 Mkm^2
IIEE[2]			0.45 Mkm^2	0.53 Mkm^2	0.18 Mkm^2	-0.07 Mkm^2
Sea ice edge position			26 km	35 km	7 km	-6 km
Sea ice edge length			10.6 Kkm	5.3 Kkm	10.6 Kkm	5.2 Kkm
First year ice		AMSR2/ SSMIS + ASCAT	1.16 Mkm^2	1.23 Mkm^2	-0.24 Mkm^2	-0.34 Mkm^2
Multi-year ice			1.02 Mkm^2	1.05 Mkm^2	-0.26 Mkm^2	-0.27 Mkm^2
			RMSD			
Sea ice drift displacement (24h)	Full domain	SAR	3.5 km	5.5 km	-1.1 km	-1.5 km
Sea ice thickness		SMOS	0.38 m	0.41 m	0.19 m	0.21 m
		CS2-SMOS[3]	0.19 m		0.001 m	
Sea ice age[4]		ESA-NERSC	105 days	110 days	-3 days	-0 days
Surface albedo[5]		OLCI	0.12	0.13	0.07	0.06

[1] Marginal ice zone (defined as the region with concentrations in the range [0.1,0.9], see IV.1.1)

[2] Integrated ice edge extent (size of ice extent mismatch between model results and observations, see IV.1.2)

[3] Comparison of modelled sea ice thickness with CS2-SMOS data are performed for 7-day averages, using model best estimates

[4] Sea ice age metrics are computed for 2019-08-01 – 2019-12-31

[5] Surface albedo metrics are computed for the summers 2020-2022

II PRODUCTION SYSTEM DESCRIPTION

Production centre's name

Nansen Environmental and Remote Sensing Center (NERSC), Bergen, Norway.

Production system's name

Arctic sea ice forecast (CMEMS name: ARCTIC_ANALYSISFORECAST_PHY_ICE_002_011)

Production centre description for the version covered by this document

The product is produced by the Arctic Monitoring and Forecasting Centre (Arctic MFC) using the stand-alone neXtSIM-F numerical prediction system for sea ice. The neXtSIM model uses an original design sea ice model that provides the best possible representation of the mechanical properties of sea ice: the rheology follows the Brittle Bingham Maxwell model (BBM) and the computational grid is a Lagrangian triangular mesh so that triangular parcels of sea ice can be followed together with their properties with minimum numerical smoothing (Rampal et al., 2019). A remeshing procedure is used to avoid overly deformed mesh elements. The present product consists of the following elements:

- Model: the stand-alone neXtSIM sea ice model (internal reference: version 1.3), developed at the Nansen Center.
- Native spatial resolution: the Lagrangian triangular mesh element size (which varies in both space and time) is approximately 7.5 km (triangle height, the side length is approximately 10 km).
- Time period: since August 2019.
- Data assimilation: nudging. Ice charts from the United States (US) National Ice Center (NIC) are used to adjust the sea ice concentration. These ice charts give the following categories: 1) open/loose ice (SIC<10%), 2) Marginal ice (10%<SIC<80%), and 3) Pack ice (SIC>80%). The total model SIC is adjusted if its classification disagrees with the USNIC ice chart classification. However, no change is made if the model classification is "marginal" while the USNIC ice chart classification is "pack". Since the sea ice model (neXtSIM) features many cracks and leads which are not represented in the ice charts, changing "marginal" to "pack" had a detrimental effect on the sea ice thickness after the summer (the ice was too thick in October when merged Cryosat-2/Soil Moisture and Ocean Salinity (CS2-SMOS) thickness observations became available). In addition, the model sea ice thickness is nudged towards that of CS2-SMOS during winter (October-April) when this product is available.
- Assimilated data: Ice charts, sea ice thickness.
- Atmospheric forcing: Data from the European Centre for Medium Range Weather Forecasts 's HRES (High Resolution) Model are used. The data is on a global regular lat/lon grid (0.1 x 0.1 degrees), with a temporal resolution of 6 h, and is interpolated onto the neXtSIM mesh during runtime.
- Lateral forcing data: None.
- Oceanic forcing: Daily-averaged sea surface temperature, salinity, mixed layer depth, and surface currents from ARC MFC's primary coupled ocean-sea ice forecast system (presently ARCTIC_ANALYSIS_FORECAST_PHYS_002_001_a) are used. These results are on a regular grid in

a stereographic projection (12.5 x 12.5 km), and are interpolated onto the neXtSIM mesh during runtime.

- Bathymetry: The ETOPO 2 arc-minute dataset is used to determine the basal stress on the ice (if any). The data are interpolated onto the neXtSIM mesh during runtime.

II.1 neXtSIM Production Cycle at NERSC

The model results are produced with the neXtSIM-F stand-alone sea ice model.

neXtSIM is developed and maintained by the Nansen Environmental and Remote Sensing Center (NERSC, http://nersc.no/) where it is run operationally to produce daily 1-day hindcasts and 9-day forecasts that are published daily in the CMEMS catalogue as the ARCTIC_ANALYSISFORECAST_PHY_ICE_002_011 product. In addition, on the first day of each month, the best estimates for the previous month are used to produce a monthly average file.

II.2 Observational Data

Assimilation is run daily before the forecast is launched, with SIC categories from ice charts and SIT being assimilated using nudging. The ice charts from the US NIC (the daily Arctic Marginal Ice Zone product from https://usicecenter.gov/Products) and the CS2-SMOS product from https://spaces.awi.de/display/CS2SMOS are used.

The assimilation is performed on the model mesh — the satellite observations, originally provided on a regular grid, are linearly interpolated onto the centres of the triangular mesh elements so that they can be directly compared with the neXtSIM prognostic variables.

Nudging employs a univariate data assimilation method: model concentrations are pulled towards the blended observations and other model variables (young and multi-year ice concentrations, volume fraction of ridged ice, ice and snow thickness) are only slightly adjusted for consistency. Where new ice is added during assimilation the sea surface temperature in the slab ocean is set to the freezing point, and is classified as young or not by extension of the young ice fraction from the existing ice mask.

III VALIDATION FRAMEWORK

The model domain for which sea ice results are available is displayed in Figure 1. Here, a sample depiction of the sea ice concentration is shown for illustrative purposes.

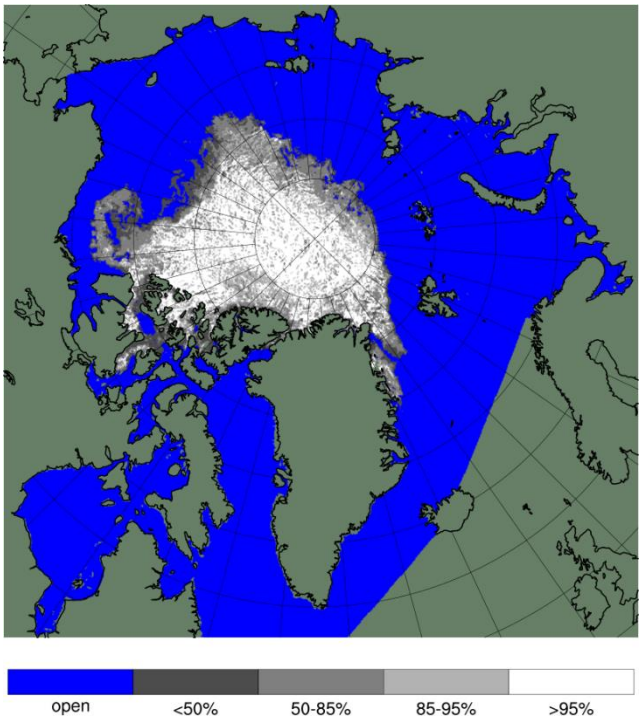

Figure 1: *Map displaying the model domain for the CMEMS implementation of the neXtSIM model. White/grey shading indicates sea ice concentration. Ice free regions are shown in blue. Displayed are model results for 2020-09-13 00:00 UTC.*

III.1 Model results

neXtSIM sea ice results are available as sets of forecast bulletins. The product that is available from the CMEMS catalogue is updated daily (see Section II.1). For the present evaluation, we completed 52 cycles of this schedule. However, for the present evaluation, model runs were extended to produce a 9-day forecast on only one of the days of the week (Fridays)..

The following model results are validated in this document:

- ✔ Sea Ice Concentration, Area, Edge Position *vs* passive microwave data;
- ✔ Sea Ice Drift *vs* synthetic aperture radar (SAR) data;
- ✔ Sea Ice Thickness *vs* surface moisture ocean salinity (SMOS) derived data;
- ✔ Sea Ice Thickness *vs* merged 7-day product from Cryosat-2 and SMOS (evaluation restricted to the model best estimate);
- ✔ Sea ice type (first year, multi-year ice) *vs* multi-sensor data;
- ✔ Sea ice age *vs* ESA Sea Ice Age CCI product based on satellite sea ice drift and concentrations;
- ✔ Albedo *vs* ocean and land colour instrument data.

Results for variables without available observational data were assessed subjectively, checking that these variables behaved in a physically sensible way. Specifically, this approach was used for assessment of

- ✔ Young ice concentration, and
- ✔ Volume fraction of ridged ice.

III.2 External operational products

Validation of the sea ice forecasts and hindcasts was performed using data from the following sources:

- Sea ice concentration, sea ice edge position:
  CMEMS product SEAICE_GLO_SEAICE_L4_NRT_OBSERVATIONS_011_001; data from the Special Sensor Microwave Imager / Sounder (SSMIS)
- Sea ice drift:
  CMEMS product SEAICE_GLO_SEAICE_L4_NRT_OBSERVATIONS_011_006; data from Synthetic-Aperture Radar (SAR) instruments.
- Sea ice thickness:
  Alfred-Wegener-Institut Soil Moisture and Ocean Salinity (SMOS) Mission product, available from ftp.awi.de:sea_ice/product/smos/nh/
  Alfred-Wegener-Institut CS2SMOS product, combining altimeter data from CryoSat-2 and SMOS data, available from ftp.awi.de:sea_ice/product/cryosat2_smos/v205/nh/
- Sea ice type:
  Ocean and Sea Ice Satellite Application Facility (OSISAF) multi-sensor product OSI-403-d, available from
  https://thredds.met.no/thredds/osisaf/osisaf_seaicetype.html; data from Advanced Microwave Scanning Radiometer 2 (AMSR2), SSMIS and the Advanced SCATterometer (ASCAT)
- Sea ice age:
  Version 2 of an advection-based sea ice age computed from climate data records of sea ice drift and concentration from OSISAF; data from SSMIS
- Surface albedo:
  Sentinel-3 Ocean and Land Colour Instrument (OLCI) instrument (Sanchez-Zapero et al, 2021), data available from https://seaice.uni-bremen.de/melt-ponds

III.3 Validation products

The validation products are:

- Sea Ice Concentration *vs* passive microwave data
  1. MAD
  2. Bias
- Sea Ice Area *vs* passive microwave data
  1. MAD, MIZ ice area

2. Bias, MIZ ice area
3. MAD MIZ extent
4. Bias, MIZ extent
5. IIEE
6. Bias, IIEE

- Ice Edge Position *vs* passive microwave data
  1. MAD, sea ice edge position
  2. Bias, sea ice edge position
  3. MAD, sea ice edge length
  4. Bias, sea ice edge length
- Sea Ice Drift *vs* synthetic aperture radar data
  1. RMSD, ice drift trajectory end point separation
  2. Bias, ice drift distance
- Sea Ice Thickness of thin ice *vs* SMOS-derived data
  1. RMSD
  2. Bias
- Sea Ice Thickness *vs* merged CS2-SMOS-derived data
  1. RMSD
  2. Bias
- Sea ice type *vs* multi-sensor data
  1. MAD, for area covered by first year ice
  2. Area bias, first year ice
  3. MAD, area with multi-year ice
  4. Area bias, multi-year ice
- Sea Ice Age *vs* ESA-NERSC data
  1. RMSD
  2. Bias
- Surface albedo *vs* OLCI data
  1. RMSD
  2. Bias

Results are provided as tabulated estimated accuracy numbers (summarized in Table 1) and time series of weekly averages after application of a three-week boxcar filter.

IV VALIDATION RESULTS

IV.1 Sea Ice Concentration, Sea Ice Area, Edge Position vs passive microwave data

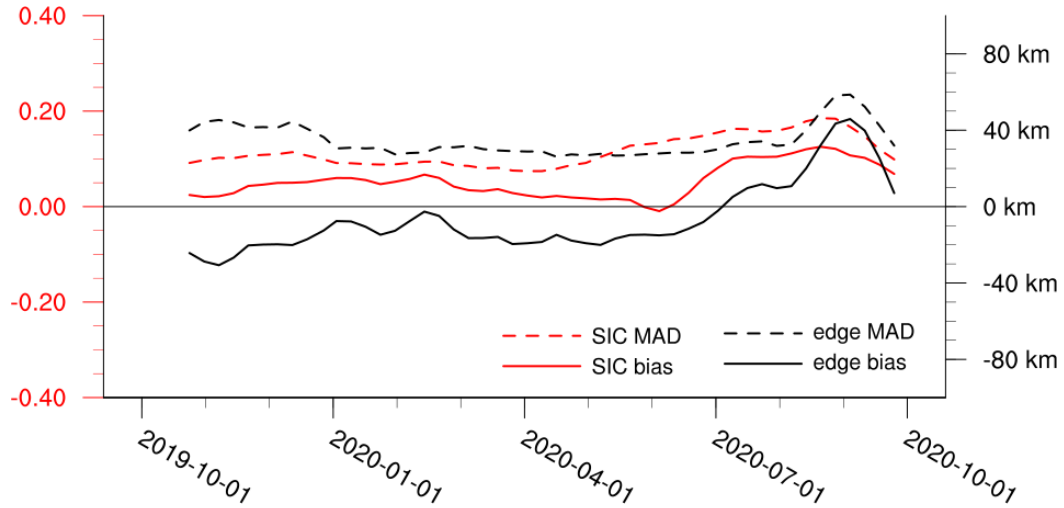

Figure 2: *Temporal evolution of sea ice concentration metrics (red lines, left axis) and ice edge position metrics (black lines, right axis, in km) for the full simulation domain and forecast day 6. Displayed are weekly results from the period 2019-10-16 – 2020-10-01, using a three-week boxcar filter.*

IV.1.1 Sea ice concentration

In Figure 2, model results for sea ice concentration (area fraction) are compared to the SSMIS passive microwave data from EUMETSAT OSISAF. These data were retrieved from CMEMS product SEAICE_GLO_SEAICE_L4_NRT_OBSERVATIONS_011_001.

The validation metrics were computed after the model results were integrated and averaged on the observation grid. Here, the marginal ice zone (MIZ) has been defined as the region with sea ice concentrations in the range [0.1, 0.9]. The analysis region for sea ice concentration metrics was limited to the observed MIZ area from observations, extended on either side of its boundary by a padding region that covers 200km of open ocean and 200km of (nearly) continuous ice cover. This eliminates artificial boosting of the quality metrics due to regions where concentrations are more or less constant and thus irrelevant.

We find that the results for sea ice concentration display only moderate deviations from the observations. Table 2 reveals that the quality deterioration with increasing forecast lead time is low, and for the sea ice concentration bias we note that the validation results suggest that model performance surprisingly improved with increasing forecast lead time.

Table 2: *Mean absolute difference (MAD) and bias (model – observation) for sea ice concentrations, based on average results for the period 2019-10-16 to 2020-10-01 for different forecast (FC) days (lead times).*

Period	2019-10-16 – 2020-10-01					
Variable	Region	Supporting observations	MAD		Bias	
			FC day 1	FC day 6	FC day 1	FC day 6
Sea ice concentration	Full domain	SSMIS	0.11	0.12	0.07	0.05

IV.1.2 Sea ice area

Sea ice area metrics are examined using the same observational data as for sea ice concentration. Three quantities are examined: the sea ice area of the marginal ice zone (MIZ ice area), the sea ice extent of the MIZ (MIZ extent), and the integrated ice edge extent (IIEE; Goessling et al., 2016). MIZ is defined as the region with sea ice concentrations in the range 0.1-0.9.

MIZ extent is defined as the total area of the MIZ. The MIZ ice area is defined as the area integral of the sea ice concentration in the MIZ.

IIEE is the area where modelled and observed ice extents does not overlap. Ice extent is defined as the area where the sea ice concentration exceeds 0.15.

Both MIZ metrics have negative biases (Table 3), i.e., on average, the observed MIZ is larger than the modelled MIZ. Furthermore, since IIEE bias is much closer to 0, the missing MIZ in the model must be on the dense ice side of the MIZ. In addition, MIZ bias and MAD are only deteriorating slightly for increasing forecast lead times.

The time series displayed in Figure 3 reveal that the IIEE mismatch is seasonally dependent. Forecast day 1 has small biases except in those months leading up to the sea ice minimum, when bias is positive. Problems with separation of melt ponds and open ocean in the passive microwave data may contribute to this bias.

Table 3: *Mean absolute difference (MAD) and bias (model – observation) for marginal ice zone (MIZ) and integrated ice edge extent (IIEE), based on average results for the period 2019-10-16 to 2020-10-01 for different forecast (FC) days (lead times).*

Period	2019-10-16 – 2020-10-01					
Variable	Region	Supporting observations	MAD		Bias	
			FC day 1	FC day 6	FC day 1	FC day 6
MIZ ice area	Full domain	SSMIS	0.69 Mkm^2	0.79 Mkm^2	-0.62 Mkm^2	-0.74 Mkm^2
MIZ extent	Full domain	SSMIS	1.02 Mkm^2	1.23 Mkm^2	-0.95 Mkm^2	-1.20 Mkm^2
IIEE	Full domain	SSMIS	0.45 Mkm^2	0.53 Mkm^2	0.18 Mkm^2	-0.07 Mkm^2

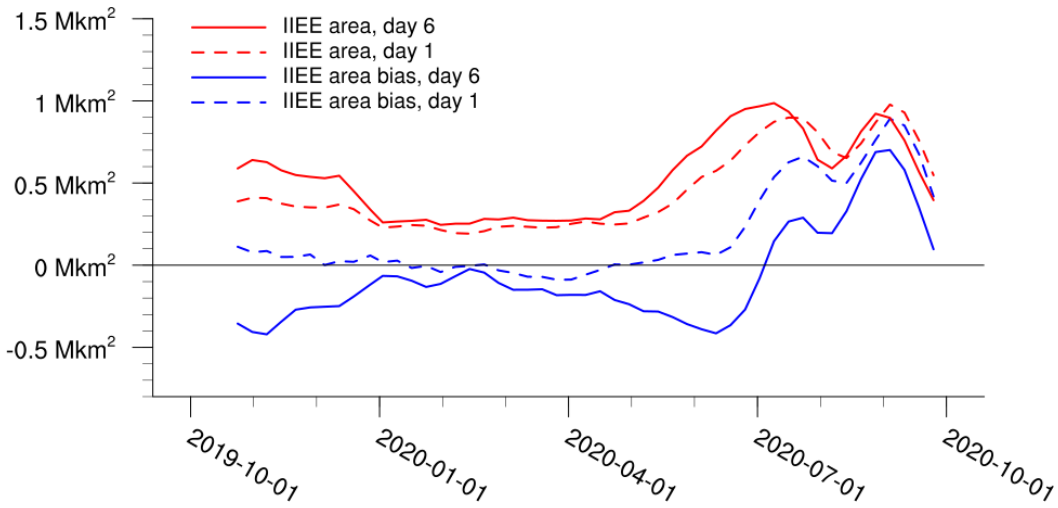

Figure 3. *Temporal evolution of the IIEE metrics. Displayed are weekly results from the period 2019-10-16 – 2020-10-01, using a three-week boxcar filter.*

MIZ biases are negative, with large discrepancies during a couple of summer months (Figure 4).

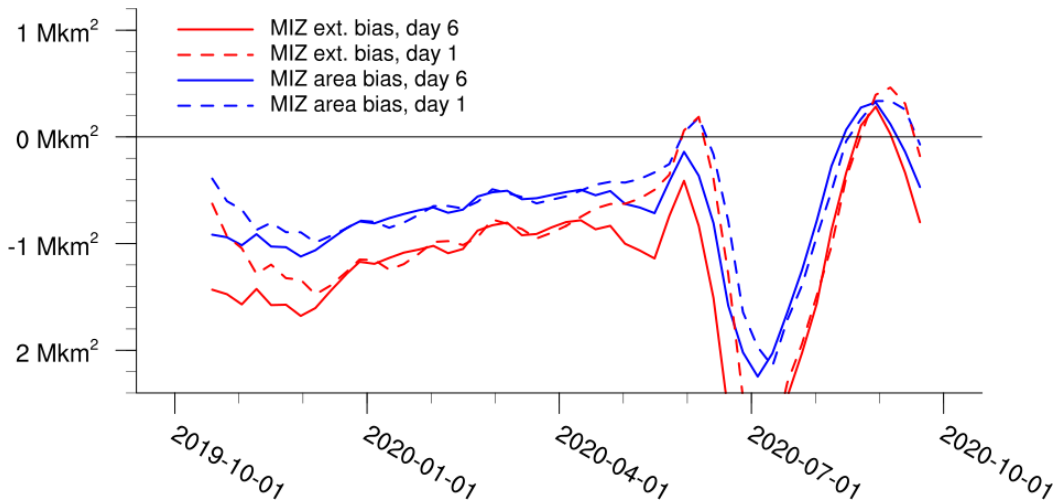

Figure 4. *Temporal evolution of the MIZ bias metric. Displayed are weekly results from the period 2019-10-16 – 2020-10-01, using a three-week boxcar filter.*

IV.1.3 Ice edge position

Ice edge position is defined as those grid cells with a sea ice concentration ≥0.15 and with a neighbouring grid cell with concentration <0.15 (see Section 2.2.1 in Melsom et al. (2019) for details). Ice edge position bias was computed according to the IIEE algorithm given by Eq. (15) in Melsom et al. (2019) applying an approach where all high resolution grid cells are accounted for upon mapping to a lower resolution grid.

Table 4: *Mean absolute difference (MAD) and bias (model – observation) for sea ice edge position, based on average results for the period 2019-10-16 – 2020-10-01. Edge bias values are taken to be positive when the model sea ice extent overshoots the observed edge.*

Period	2019-10-16 – 2020-10-01					
Variable	Region	Supporting observations	MAD		Bias	
			FC day 1	FC day 6	FC day 1	FC day 6
Sea ice edge position	Full domain	SSMIS	26 km	35 km	7 km	-6 km
Sea ice edge length	Full domain	SSMIS	10.6 Kkm	5.3 Kkm	10.6 Kkm	5.2 Kkm

While the length of the observed ice edge only varies modestly, the length of the model ice edge has a large temporal variability (Figure 5).

A further examination reveals that during summer, the inflated model edge length is associated with a large number of polynyas and openings resulting from melting along coastlines.

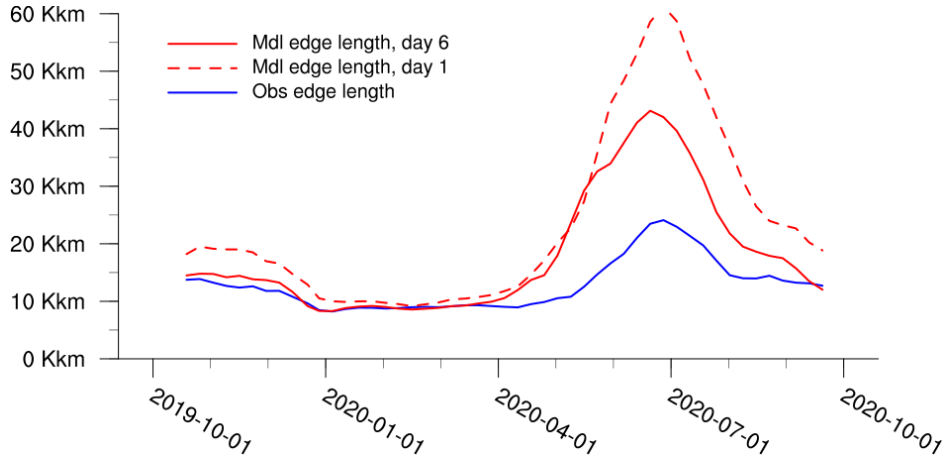

Figure 5. *Temporal evolution of ice edge length. Displayed are weekly results from the period 2019-10-16 – 2020-10-01, using a three-week boxcar filter.*

IV.2 Sea Ice Drift vs synthetic aperture radar data

Model results for displacement due to sea ice drift have been compared to displacement data derived from Synthetic Aperture Radar imagery. These data were retrieved from CMEMS product SEAICE_GLO_SEAICE_L4_NRT_OBSERVATIONS_011_006.

In order to produce a set of comparable observations and model results, 24 h trajectories were first derived from integration of the hourly model results for sea ice drift velocity. Then, the observational displacement data, which are typically available at temporal resolutions of 18-24 h, were normalized to 24 h, assigning weights according to how much of the drift that took place during the two dates for which the displacement data usually were taken from (e.g. observed drift from noon to noon on consecutive dates were given the same weight when comparing to modelled drift for the two days).

Here, we present results for two validation quantities, the bias in the 24 h displacement distance and the root-mean-square (RMS) of the separation distance between the vector end points (observations were extrapolated for any remaining time in order to represent 24 h drift). In Figure 6 we provide an illustration that explains how we approach validation of vector properties. Metrics for such properties are not used frequently for quality assessment.

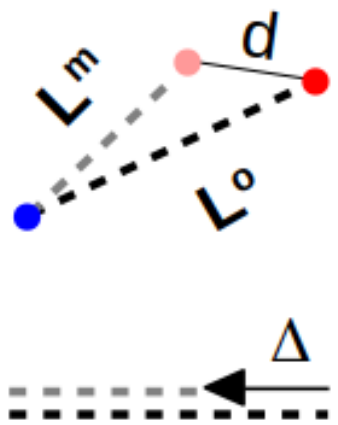

Figure 6: *Illustration of the drift quality metrics. The blue dot marks the initial position from which drift is considered. Red and pink dots indicate the observed and modelled end position of drift for a given time period (here one day), respectively. The separation distance, d, is the offset between the modelled and observed end positions. Drift length bias (Δ) is indicated by the arrow near the bottom. It is positive when the model displacement (L^m) is greater than the observed displacement (L^o). Thus, bias is positive when the arrow points to the right and vice versa. Another interpretation of* d *is that the model end position is somewhere on a circle with a radius d, centred on the red marker. This implies a restriction on the angle between lines* L^o *and* L^m.

RMS of the separation distance for the 24 h sea ice vector drift displacements indicates a moderate deterioration with increasing forecast lead time (Table 5). Furthermore, there is a negative bias (Δ in Figure 6 is negative).

Since the RMS of the separation distance is smaller than the average trajectory length from observations, we can infer that model trajectories point in the same general direction as observations. Moreover, since the RMS of the separation distance is significantly smaller and close to half the displacement during the 24 h drift, we conclude that directional accuracy is significantly higher for shorter forecast lead times.

Table 5: *Root mean square difference (RMSD) and bias (model – observation) for sea ice displacement due to 24 h drift. Results are given for the period 2019-10-16 – 2020-10-01.*

Period	**2019-10-16 – 2020-10-01**						
Variable	**Region**	**Supporting observations**	**RMSD**		**Bias**		**Mean observed trajectory length**
			FC day 1	**FC day 6**	**FC day 1**	**FC day 6**	
Sea ice drift displacement (24 h)	Full domain	SAR	3.5 km	5.5 km	-1.1 km	-1.5 km	7.1 km

There are no obvious seasonal patterns in quality (Figure 7). If we let the listed metrics in Table 5 represent individual vectors, the directional offset between observations and model end positions would be 29° and 50° for forecast days 1 and 6, respectively.

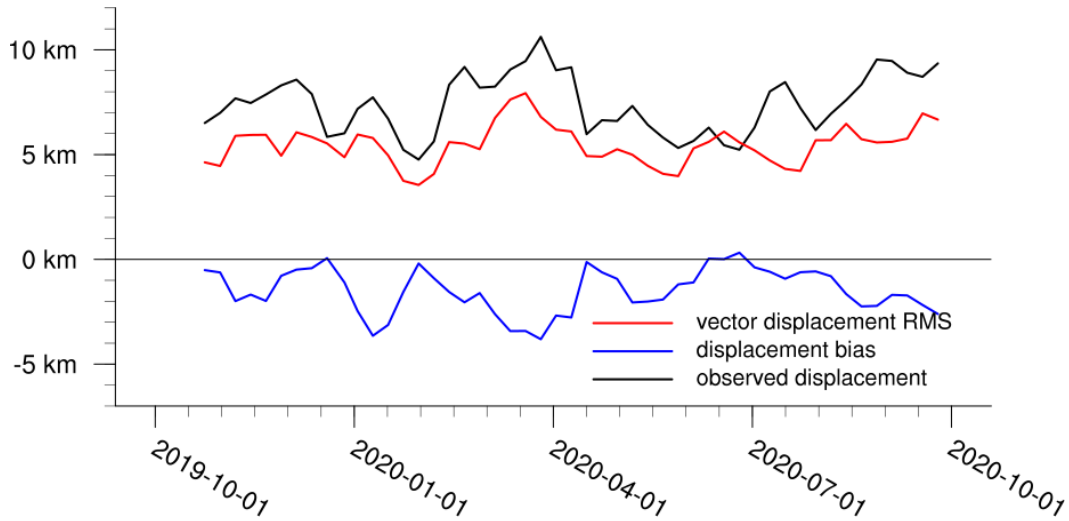

Figure 7: *Temporal evolution of 24h sea ice drift displacement metrics for the full simulation domain, for forecast day 6. Displayed are weekly results from the period 2019-10-16 – 2020-10-01, using a three-week boxcar filter. A displacement bias of 1.7 km/day corresponds to a difference of 0.02 m/s.*

IV.3 Sea Ice Thickness vs SMOS-derived data

Model results for sea ice thickness have been compared to data from the SMOS satellite mission. The SMOS data were downloaded from the Alfred-Wegener-Institut. Model results were interpolated onto the observational grid. Note that the SMOS data represent ice thickness of about 1 m or less (i.e., thin ice). Regions where the ice was too thick or where other quality issues could be identified have been discarded from the analysis. The resulting masking was then applied to the model results, such that the spatial data coverage was time dependent. Note also that SMOS data is only available for the period 15 October – 14 April.

We present results for sea ice thickness bias and RMSD for those regions where observational data was available.

Table 6: *Root mean square difference (RMSD) and bias (model – observation) for sea ice thickness in regions with thin ice. The results are based on averages from available data during the period 2019-10-15 – 2020-04-14.*

Period	**2019-10-15 – 2020-04-14**					
Variable	**Region**	**Supporting observations**	**RMSD**		**Bias**	
			FC day 1	**FC day 6**	**FC day 1**	**FC day 6**
Sea ice thickness	Full domain	SMOS	0.38 m	0.41 m	0.19 m	0.21 m

The metrics deteriorated only to a modest degree for increasing lead times (Table 6), possibly related to the slowness in sea ice thickness changes due to freezing and melting, which are processes that act over longer time scales. For ice covers ≤ 1m, the discrepancies are moderate (Table 6), with the model overestimating sea ice thickness.

To examine this more closely, we divided ice thickness into four different categories and conducted the comparison based on these categories (Table 7). For all categories, the distribution of model results is skewed toward thicker ice. However, the fraction of the model category with thickness >1 m increases with increasing observed sea ice thickness (rightmost table column). This suggests that the model sea ice thickness results, while showing a strong positive bias, have a geographical distribution that, to a certain degree, matches the geographical distribution of the observed variations in thickness. In other words, where the modelled sea ice thickness is >1 m, the observed thickness is generally close to 1 m (in areas for which observations are available).

Table 7: *Model vs observed sea ice thickness category distributions for regions with thin ice. The results are based on available data during the period 2019-10-16 – 2020-10-01. With the exception of the rightmost column, values in the table show fractions of all results. Grids with no ice ('Open ocean') in both products were discarded prior to performing the analysis. Tabulated fractions are for the day 6 of the forecasts.*

Period:	**2019-10-16 – 2020-04-14**						
		Thickness category - model					***Fraction, >1 m / all***
Thickness category - observations		**Open ocean**	**<0.3 m**	**0.3-0.6 m**	**0.6-1 m**	**>1 m**	
	Open ocean		0.020	0.003	0.001	0.000	*0.01*
	<0.3 m	0.105	0.151	0.089	0.053	0.027	*0.06*
	0.3-0.6 m	0.002	0.034	0.087	0.099	0.065	*0.23*
	0.6-1 m	0.001	0.008	0.041	0.103	0.110	*0.42*
	>1 m	0.000	0.000	0.000	0.000	0.000	*N/A*

The results from Table 7 are summarized in the bar chart in Figure 8.

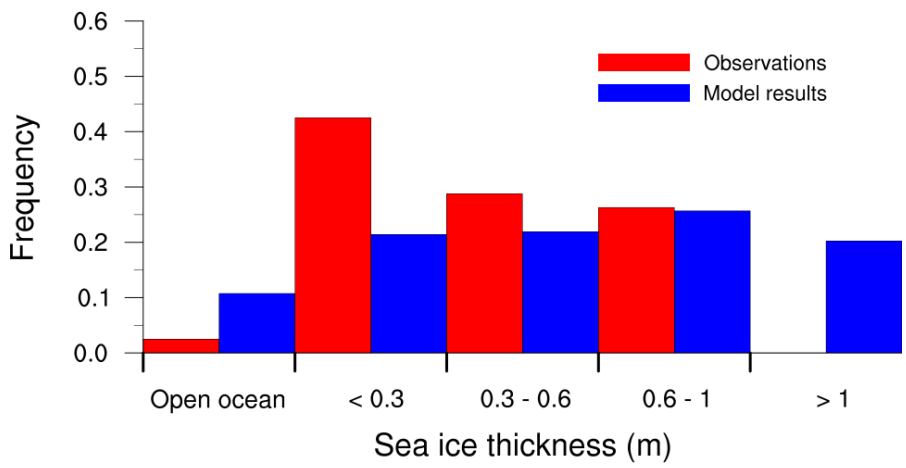

Figure 8: *Sea ice thickness category distribution in the region with valid SMOS data for forecast day 6. See also Table 7.*

There is a seasonal variability in the quality of sea ice thickness forecasts in regions with thin ice (Figure 9). Close to the sea ice minimum, differences are relatively small, while discrepancies become fairly significant around the time of the sea ice maximum. Furthermore, both bias and RMSD show a smooth temporal evolution. This indicates that changes are driven by processes acting over relatively long time

scales, e.g., in processes such as anomalous melting and freezing due to heat flux biases, either between sea ice and atmosphere or, more likely, in ocean-sea ice fluxes. Another conceivable contributing factor is the persistent underrepresentation of divergence in the model results in regions covered by SMOS data, which would also lead to a positive bias. Finally, we note that the domain-wide bias is positive for all simulation dates. The relevant overall spatial distribution of category match-ups in Table 7 has 20% of grids below the diagonal, while the corresponding values above the diagonal is 49%.

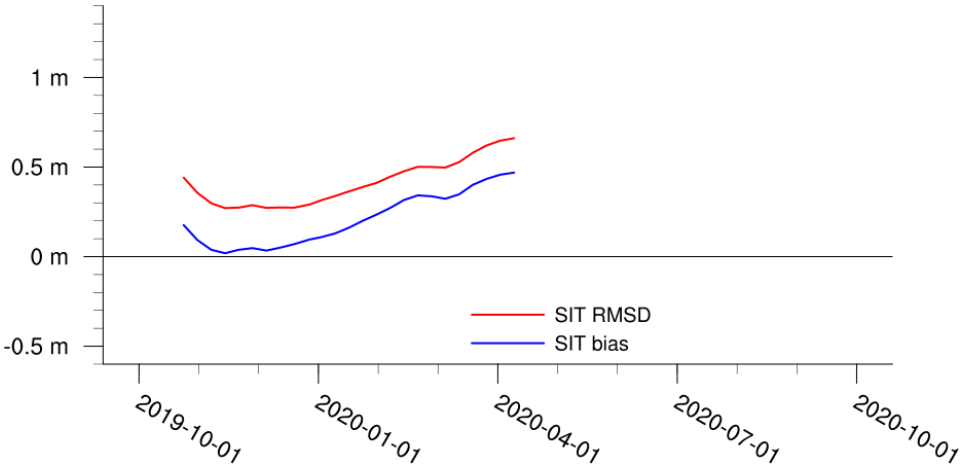

Figure 9: *Temporal evolution of sea ice thickness metrics for the full simulation domain, for the day 6 of the forecasts. Displayed are weekly results, using a three-week boxcar filter. The period with results is 2019-10-16 – 2020-04-16, restricted by availability of satellite data.*

IV.4 Sea Ice Thickness vs CS2-SMOS data

Model results for sea ice thickness have been compared to merged data (CS2SMOS) from the Cryosat-2 and SMOS satellite missions, downloaded from the Alfred-Wegener-Institut. The model results were interpolated onto the observational grid and averaged with a 7-day moving window, since the CS2-SMOS data represent collated and optimally-interpolated observations over 7-day intervals. Note also that CS2-SMOS data are only available from mid-autumn to early spring (15 October – 14 April).

For simplicity, given the 7-day averaging period for the observations, we use the best estimate for the model-data comparison . This should not affect forecast results as ice thickness is a slowly-varying variable.

At the start of the winter, the model significantly (by >0.3m) overestimates sea ice thickness (Figure 10, right panel). However, as data continues to be assimilated by the model, the bias roughly remains between -0.1 and 0.1 m for the remaining period and within the bounds of observational uncertainty (shaded area). Initially, also the forecast RMSD is high compared to observational RMSD, but the model value also drops quickly and approaches its observational counterpart as the data assimilation starts to take effect. Table 8 summarises Figure 10.

Table 8: *Root mean square difference (RMSD) and bias (model – observation) for sea ice thickness. The results are based on averages from available data during the period 2020-10-16 – 2021-04-14.*

Period	**2020-10-16 – 2021-04-14**			
Variable	**Region**	**Supporting observations**	**RMSD**	**Bias**
Sea ice thickness	Full domain	CS2-SMOS	0.19 m	0.001 m

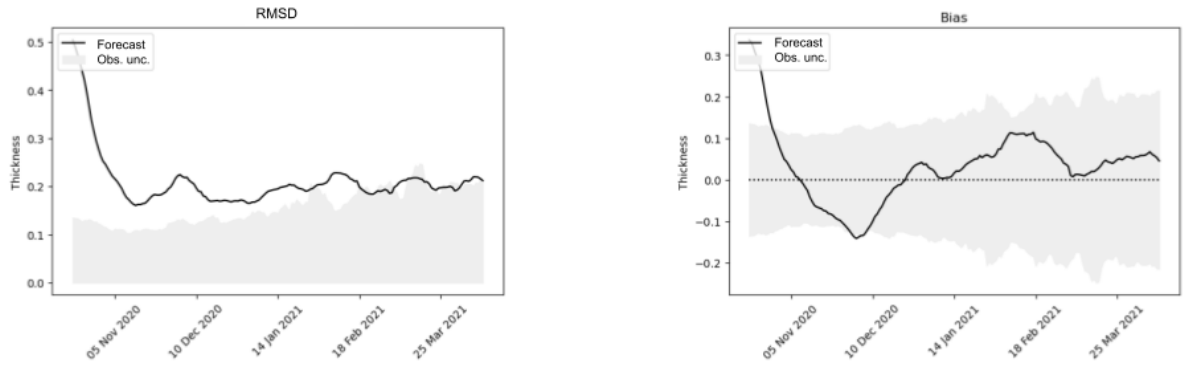

Figure 10: *Temporal evolution of the sea ice thickness metrics for the full simulation domain, for the best estimate. The period with results is 2020-10-16 – 2021-04-16, restricted by availability of satellite data.*

IV.5 Sea Ice Type vs multi-sensor analysis

Sea ice type differentiates between ice that has frozen after the latest sea ice minimum ("first year ice"), and sea ice that survived the minimum ("multi-year ice"). Model results for sea ice type have been compared to a multi-sensor analysis based on data from passive microwave and scatterometer sensors. The present analysis uses data from product OSI-403-d from the EUMETSAT Ocean and Sea Ice Satellite Application Facility (OSISAF). Note that these data are only available for the period 1 October – 10 May.

We present results for the sea ice type bias and mean absolute differences (MAD), for those regions where reliable data was available (regions where data was flagged as "low confidence" were excluded from the analysis).

Changes in sea ice type can mostly be attributed to melting and freezing, which are processes that take place on longer time scales than about one week. Hence, it is not surprising that the metrics seem nearly unaffected by forecast lead time (Table 9).

Table 9: *Mean absolute difference (MAD) and bias (model – observation) for sea ice type. The results are based on averages from available data during the period 2019-10-16 – 2020-05-10.*

Period	2019-10-16 – 2020-05-10					
Variable	Region	Supporting observations	MAD		Bias	
			FC day 1	FC day 6	FC day 1	FC day 6
First year ice	Full domain	AMSR2/SSMIS + ASCAT	1.16 Mkm^2	1.23 Mkm^2	-0.24 Mkm^2	-0.34 Mkm^2
Multi-year ice	Full domain	AMSR2/SSMIS + ASCAT	1.02 Mkm^2	1.05 Mkm^2	-0.26 Mkm^2	-0.27 Mkm^2

Looking at the temporal evolution of metrics, we note that MAD is slightly lower during the first two months after the time of the sea ice minimum (mid-September), while remaining fairly constant after November and with similar values for first- and multi-year ice (Figure 11).

For most of the investigated time period, biases for first- and multi-year ice are out of phase, with a surplus of first year ice around new year, and the reverse situation from early April onwards.

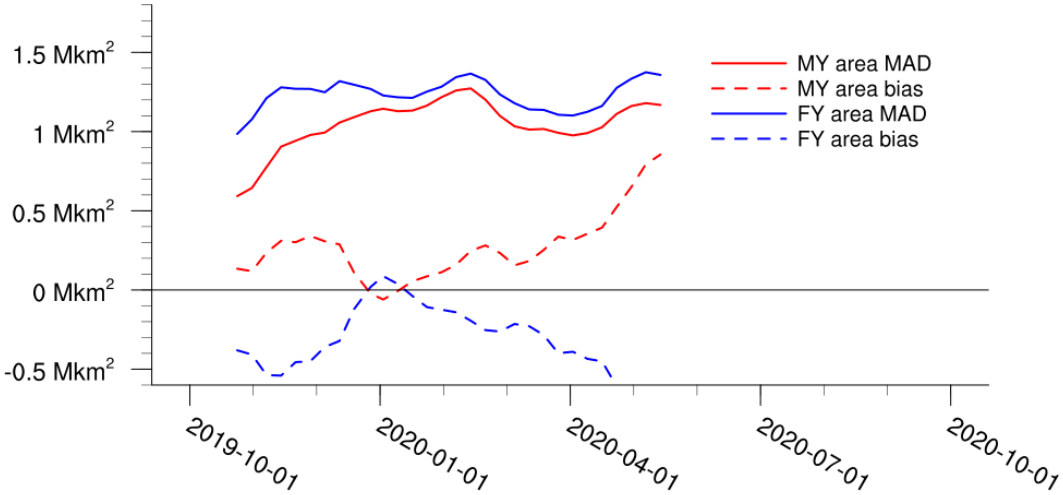

Figure 11: *Temporal evolution of the sea ice type metrics for first-year (FY) and multi-year (MY) ice and forecast day 6. Displayed are weekly results, using a three-week boxcar filter. The period with results is 2019-10-16 – 2020-05-15, restricted by availability of satellite data.*

IV.6 Sea ice age vs ESA-NERSC product

The satellite-derived NERSC sea ice age product is an evolution of the ESA Sea Ice Age product described in Korosov et al. (2018). It uses sea ice drift and concentrations to calculate ice age distribution at every location, discretized on a Lagrangian sea ice mesh. The overlap period between the observation product and the reforecast period is 2019-08-01 to 2019-12-31.

There is a fairly good agreement between the model and observations over this period, with the bias oscillating between ±30 days (Figure 12) but having a mean very close to zero (Table 10). RMSD is between 70 and 140 days (about 105 days on average, see Table 10), higher in August and September before dropping over the course of October to a lower level that persists through November and December. The comparison is fairly independent of forecast lead time.

Table 10: *MAD and bias (in days) for sea ice age, based on results for August-December 2019 for different forecast (FC) days (lead times).*

Period	2019-08-01 – 2019-12-31					
Variable	Region	Supporting observations	RMSD		Bias	
			FC day 1	FC day 6	FC day 1	FC day 6
Sea ice age	Arctic Ocean	ESA-NERSC sea ice age product	105	110	-3	-0

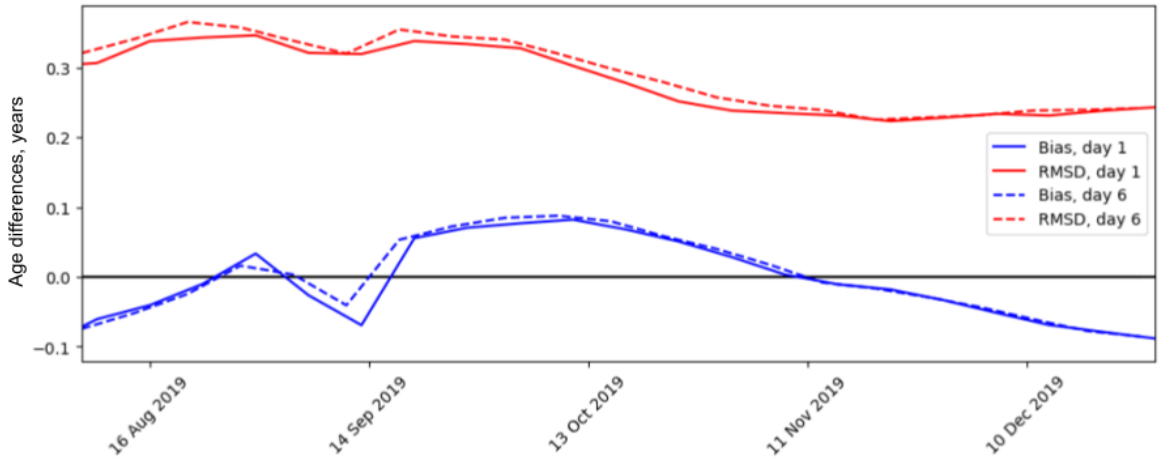

Figure 12: *Time evolution of sea ice age metrics (in years) for August-December 2019, for the full domain and forecast lead times of 1 and 6 days.*

IV.7 Surface albedo vs OLCI data

Modelled surface albedo for the summers of 2020-2022 was compared to OLCI-derived broad-band albedo (https://seaice.uni-bremen.de/melt-ponds). The model results were interpolated onto the 12.5-km polar-stereographic observation grid and bias and RMSD were calculated where observations were available. The model exhibits a consistently positive biased with values ranging from 0 to 0.1. RMSD is between 0.1 and 0.175 (Figure 13). The agreement is generally better at the beginning of summer (May to mid-June), with greater differences later on. However, in 2021 the agreement improves as the summer progresses. There is little variability in the accuracy with the forecast lead time (Table 11).

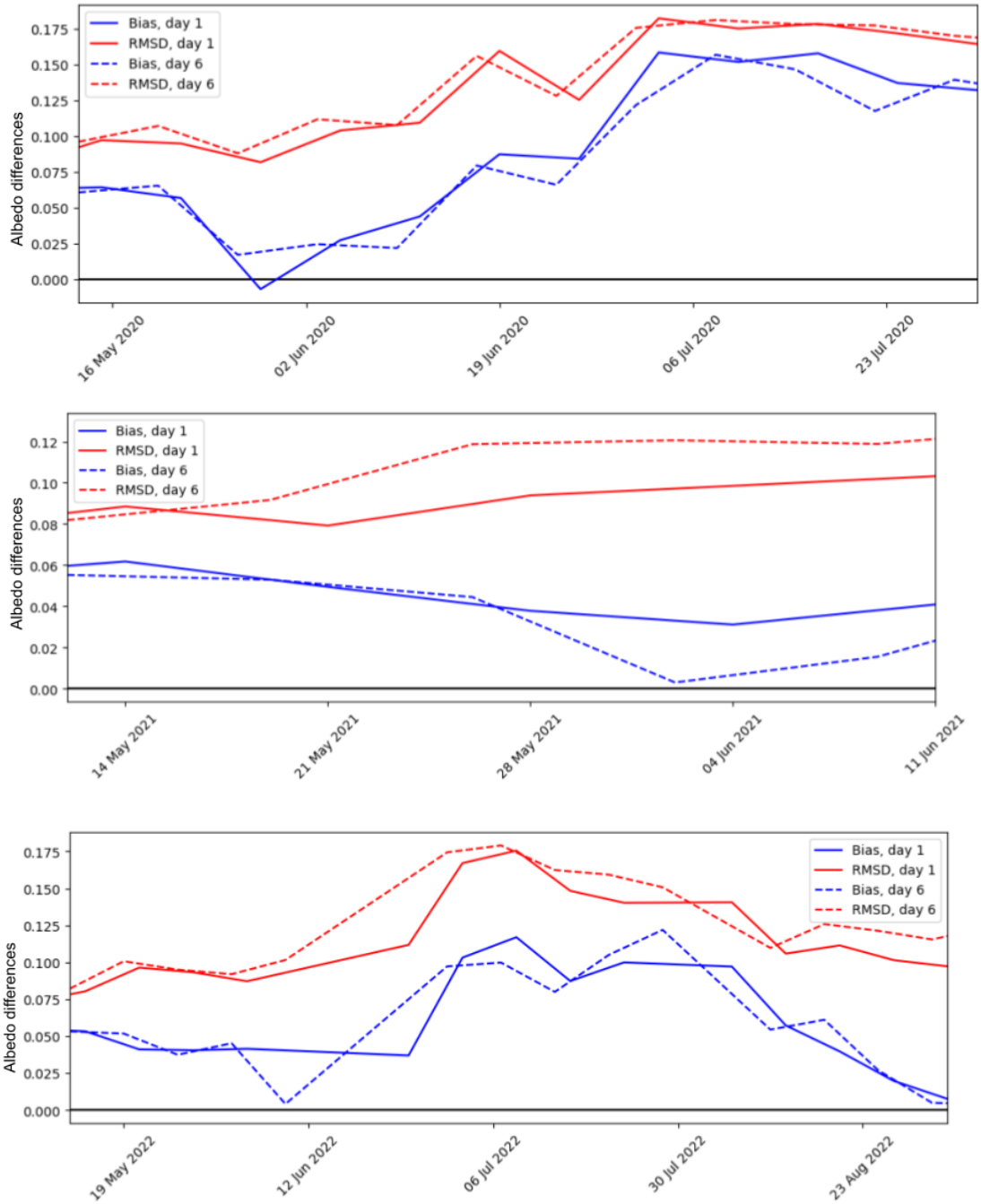

Figure 13: *Time evolution of surface albedo accuracy metrics for the summers of 2020-2022, for the full domain (where observations were present), and for forecast lead times of 1 and 6 days.*

Note that the observation product has changed since v1.3 of this document, where the modelled albedo (for 2020) was found to have a negative bias of <-0.1. This change of sign is due not just to the modelled albedo increasing, but also to the albedo estimated from observations decreasing after a new version of the U. Bremen product was released.

Table 11: *Accuracy numbers for surface albedo based on results for the summers of 2020-2022 with forecasts.*

Period	2020-05 – 2022-07					
Variable	Region	Supporting observations	RMSD		Bias	
			FC day 1	FC day 6	FC day 1	FC day 6
Surface albedo	Full domain	OLCI	0.12	0.13	0.07	0.06

IV.8 Sea ice area fraction of young ice: visual inspection

Since this variable is difficult to validate through observations, due to the scarcity of observational data, we performed a qualitative (visual) validation to check it is behaving in a sensible manner. In Figure 14, we show maps for months when young ice is being produced (it is rarely produced in June, July or August) . In those months when young ice is present, it almost always occurs near the ice edge. If ice has been added during assimilation, either all, some, or none of the new SIC may be designated as young ice, by extending the young ice fraction from the pre-existing ice mask. This results in some uncertainty regarding the concentration of young ice near the ice edge.

Young ice is also produced to varying degrees in openings (cracks/leads) in the pack. This is especially pronounced in the first few freezing months (September to November) when the total SIC is lower, although it is still present later in the freezing season to some extent, when the wind is strong enough to cause openings to form in the pack.

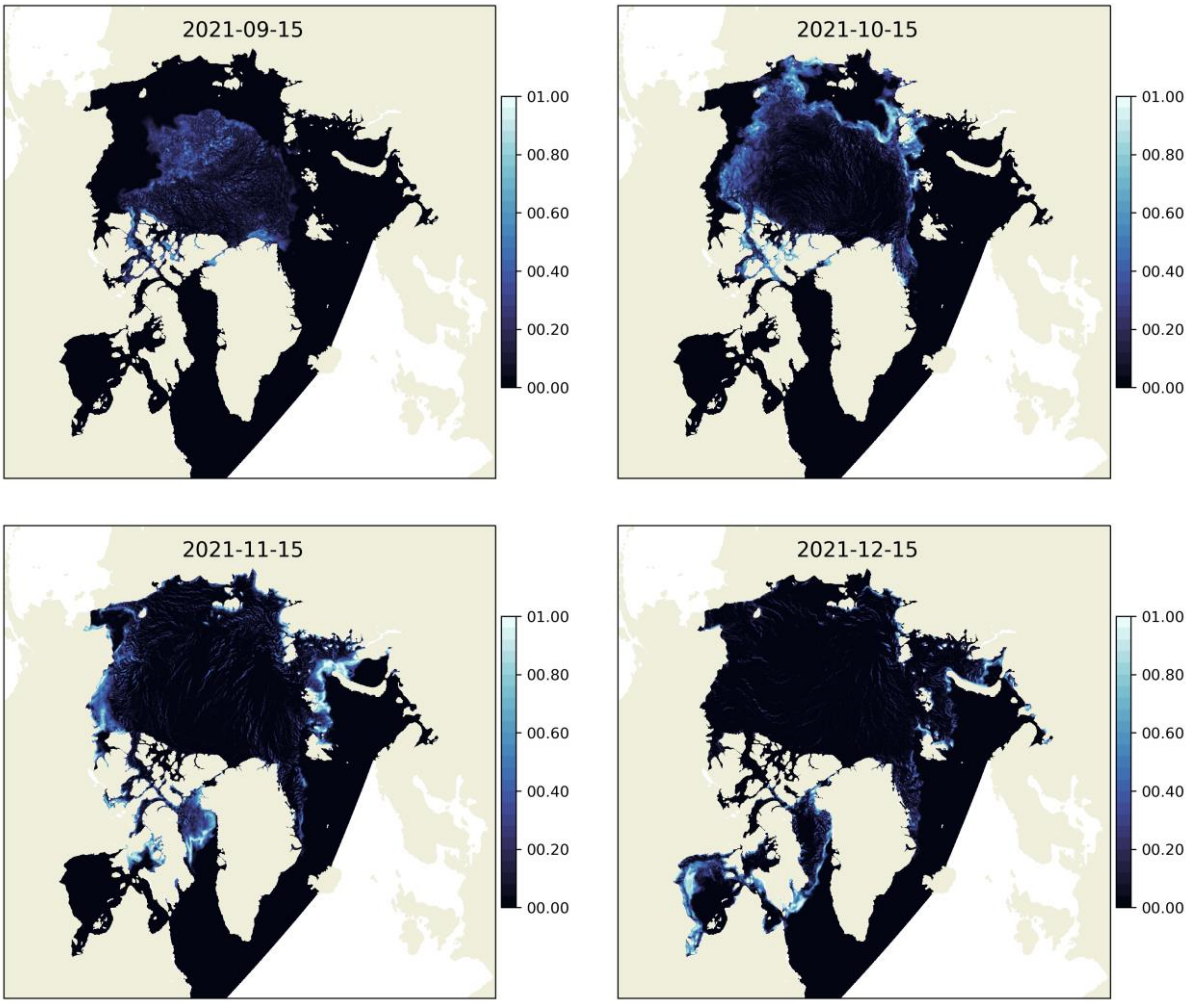

Figure 14: *Maps of model results for daily means of the variable "sea ice area fraction of young ice" for the indicated dates.*

IV.9 Sea ice volume fraction of ridged ice: visual inspection

Again, this variable is difficult to validate with observations, due to the scarcity of reliable observations which is why we employed a qualitative (visual) validation using individual maps from different periods of the year (from October 2021 to July 2022) (Figure 15). Throughout the year, there is a high amount of ridging near coastlines. This is true even in regions with first-year ice such as off the coast of Siberia, in Hudson Bay, and in the Labrador Sea. Towards the end of winter (April 2022 map in Figure 16), the Hudson Strait is very highly ridged.

Far from coasts, multi-year ice (MYI) has a ridge fraction of about 0.4, although there is some spatial variability (MYI extent is approximately the same as the ice extent shown in the map of young ice for 15 September 2021 in Figure 14). A value of 0.4 means that within a few kilometers of sea ice about 40% of the total volume of sea ice is ridged, which may seem high compared to what is visible from the deck of an icebreaker. In October, FYI has only begun to ridge near the New Siberian Islands. By January, FYI

has begun to show some signs of ridging away from coasts, but the level is clearly lower than for MYI. By April, a little more ridging has appeared in FYI but there are still patches of low ridge fraction. By July, there is even more ridging in FYI regions, although still slightly less than in MYI.

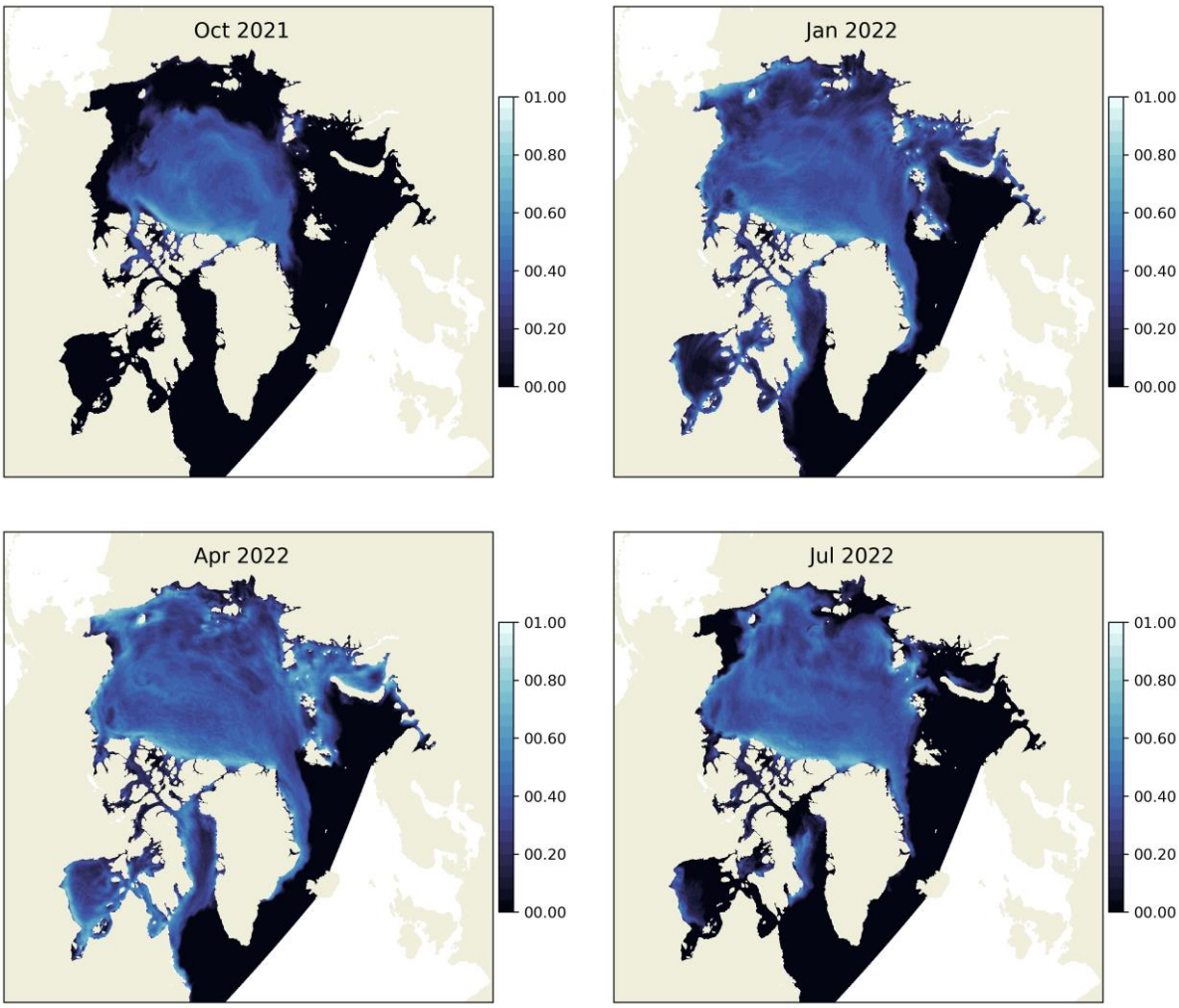

Figure 15: *Maps of monthly means of the variable "sea ice volume fraction of ridged ice" for the indicated dates.*

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This document describes the fifth version of the Arctic MFC sea ice forecasting system.

Table 12: *List of events, outages and/or changes for each version of the present product.*

Date	Change/Event description	System version	other
7/7 2019	First version of the neXtSIM-F forecast using the Maxwell-Elasto-Brittle (MEB) rheology.	1.0	
17/12 2020	Upgraded sea ice rheology from the MEB rheology to the Brittle Bingham-Maxwell rheology (BBM).	1.1	
17/12 2021	Improvements to the BBM sea ice rheology were added. Better tuning of dynamic and thermodynamic model parameters, and in the assimilation parameters. Extended model domain further south to cover the Labrador Sea, Hudson Bay and the St. Lawrence Estuary.	1.2	
31/8 2022	The assimilation now uses daily ice charts ("Daily Marginal Ice Zone") to give improved initial ice extent, particularly near the coasts, where the OSISAF SSMIS signal is affected by the presence of land. Two new variables have been added to the model output: sea ice albedo and sea ice area fraction of multi-year ice. The dynamic forcing now uses ocean currents from the 30m level. Further tuning has been performed of dynamic and thermodynamic parameters to improve quality.	1.3	
9/8 2023	The assimilation now includes nudging of the sea ice thickness towards the CS2-SMOS product. Three new variables have been added to the model output: sea ice age, sea ice area fraction of young ice, and sea ice volume fraction of ridged ice. Further tuning has been performed of thermodynamic parameters to improve quality. In particular, the systematic underestimation of the sea ice albedo has been improved.	1.4	

VI QUALITY CHANGES SINCE PREVIOUS VERSION

August 2023:

The new version of this product (v1.4) includes results for three new variables: the concentration of young sea ice, sea ice age, and the volume fraction of ridged sea ice. Validations results for these quantities are provided in Sections IV.8, IV.6, and IV.9, respectively.

Albedo quality metrics show an improvement by about 50%. Furthermore, bias for sea ice thickness, based on a comparison with SMOS data, is reduced by approximately 20%. Results for sea ice edge position are also improved for the longer (6-day) forecast ranges.

In contrast, some results have deteriorated compared to the previous version, although to a much smaller degree than the major improvements mentioned above. These slight deteriorations are found for sea ice concentration, and in the biases for areas of first year ice and multi-year ice.

VII REFERENCES

Goessling, H. F., Tietsche, S., Day, J. J., Hawkins, E., and Jung, T.: Predictability of the Arctic sea ice edge, Geophys. Res. Lett., 43, 1642-1650, https://doi.org/10.1002/2015GL067232, 2016.

Korosov, A. A., Rampal, P., Pedersen, L. T., Saldo, R., Ye, Y., Heygster, G., Lavergne, T., Aaboe, S., and Girard-Ardhuin, F.: A new tracking algorithm for sea ice age distribution estimation, The Cryosphere, 12, 2073–2085, https://doi.org/10.5194/tc-12-2073-2018, 2018.

Melsom, A., Palerme, C., and Müller, M.: Validation metrics for ice edge position forecasts, Ocean Sci., 15, 615-630, https://doi.org/10.5194/os-15-615-2019, 2019.

Rampal, P., Dansereau, V., Olason, E., Bouillon, S., Williams, T., Korosov, A., and Samaké, A.: On the multi-fractal scaling properties of sea ice deformation, The Cryosphere, 13, 2457–2474, https://doi.org/10.5194/tc-13-2457-2019, 2019.

Sánchez-Zapero. J., and 8 co-authors: Prototype for Surface Albedo Retrieval Based on Sentinel-3 OLCI and SLSTR Data in the Framework of Copernicus Climate Change, 2021 IEEE International Geoscience and Remote Sensing Symposium IGARSS, 2377-2380, https://doi.org/10.1109/IGARSS47720.2021.9555099, 2021.